

Patterns to Principles

Theory Construction in the Social Sciences

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General Information

Course name	From Patterns to Principles: Theory Construction in the Social Sciences
Instruction language	English
Type of instruction	Interactive lectures and group work
Level	Excellence Program TSB
Course load	6 ECTS
Lectures	Dr. C. J. Van Lissa

This course combines literature from empirical social science, methods and statistics, and philosophy to offer a multidisciplinary perspective on theory in the social sciences. There is no textbook: students will read and present on primary literature and discuss it in class. The course uses challenge based learning and a flipped classroom. While default challenges and corresponding materials are available, students are encouraged to bring their own, thereby connecting this course to their other coursework (or future thesis). The course relies on group work and explicitly allows for specialization (i.e., not all students need to master all course-relevant skills).

Learning objectives

This course introduces students to the foundations and applications of theory in social science research. We will analyze the need for better theories based on the current “theory crisis” in empirical social science. We define the characteristics and affordances of “good theories”: what do they look like, and how do we use them in (empirical) research? Students will engage with theory construction methods to formalize existing theories or construct their own, and are introduced to formal modeling and simulation studies as tools for theory development and testing. By the end of the course, students will be able to critically assess existing theories and their usage in empirical articles, construct theory, simulate theory-implied data, use theory in empirical research, and relate critically to the state of theory in the social sciences.

Skills

After taking this course, students are able to:

- Remember relevant constructs from the literature
- Understand theory construction methods and their philosophical pitfalls
- Apply theory in empirical research
- Create new - or formalize existing - theory
- Evaluate the use of theory in empirical research

Study Load

This is the breakdown of the expected study load:

Activity	Duration	Times	Total
Synchronous			
Lectures	2.0	14	28.000000
Tutorials	2.0	14	28.000000
Asynchronous			
Reading	1.0	14	14.000000
Assessment type(s)			
Portfolio	2.0	14	30.000000
Poster	6.0	1	6.000000
Exam	1.5	2	3.000000
Total			109.000000
ECTS			3.892857

Assessment

Your grade for the course is based entirely on your “portfolio”, comprised of the following components:

Component	Type	Percentage	Description
Multiple choice exam	individual	30.00%	During 11 lecture sessions, a 4-question multi
Assignment	individual	10.00%	Write a one-paragraph (max 500 word) summ
Presentation of theory specification	group	20.00%	Present your group project to class
Final theory report	group	40.00%	Write a report about your final theory, and a

Group Work

For this course, we want to stimulate you to engage with the material, with one another, and with the teachers. Therefore, some portfolio assignments are made in groups. There are aspects of learning in groups that can really improve your knowledge, like discussion and peer feedback. Groups comprise 3-4 members and are created during the first tutorial session (Section [1.2.1](#)). We encourage students to reflect on their [roles in group work](#), and to explicitly divide roles before each session or switch roles each week. At minimum, it's useful to always have one Facilitator. If your group tends to be unanimous, appoint a Devil's advocate. If your group tends towards disagreement, appoint an Arbitrator.

Course materials

You do not need a book for this course. This GitBook provides all relevant course materials, along with primary literature.

Course schedule

Always check the latest course schedule on [TimeEdit](#). For an overview of the content, see below:

Dates	Activities	Topic
24/08 28/09	HC, LA	Why theory needs an update: Theory Crisis
31/08 04/09	HC, LA	Why theory needs an update: What Makes a Good Theory?
07/09 11/09	HC, LA	Why theory needs an update: Strategic Ambiguity
14/09 18/09	HC, LA	How to Construct Theory: Latent Theory in Published Papers
21/09 25/09	HC, LA	How to Construct Theory: Formal Theory Specification
28/09 04/10	Conference	Theory Methods Conference
05/10 9/10	HC, LA	How to Construct Theory: Analogical Modeling
12/10 16/10	Tentamens blok 1	
19/10 23/10	HC, LA	How to Construct Theory: Simulation Studies
26/10 30/10	HC, LA	How to Construct Theory: Agent Based Modeling
02/11 06/11	HC, LA	What is Next: Machine Learning-Informed Theory Construction
9/11 13/11	HC, LA	What is Next: Causal Discovery
16/11 20/11	HC, LA	What is Next: FAIR Theory
23/11 27/11	HC, LA	Building Theory: Student Theory Workshop
30/11 04/12	HC, LA	Building Theory: Student Theory Workshop
07//12 11/12	Tentamens blok 2	
14/12 18/12	Tentamens blok 2	
21/12 25/12	Vakantie	
28/12 01/01	Vakantie/Studieweek	
04/01 08/01	Hertentamens blok 1 en 2	
11/01 15/01	Hertentamens blok 1 en 2	
18/01 22/01	Hertentamens blok 1 en 2	

Communication

Communication about the course occurs through Canvas (Login with your student ID and password).

Staff

Coordinator/lecturer:

[dr. Caspar J. Van Lissa](#)

Lab sessions

TBD

Contact hours

This course consists of 14 lectures and 11 lab sessions (LA).

The course consists of a common part and a major specific part. The first 10 lectures are common lectures, which means they have to be followed by students from all majors (i.e., business and economics, cognitive neuroscience and social sciences). The last four lectures are major specific.

During the lab sessions you will work on applied exercises supervised by a teaching-assistant. These exercises are always related to topics discussed during the lectures. The assignments of the SPSS lab sessions are available via Canvas. You will be able to ask your lab teacher questions during the lab sessions.

Software

The official software for this course is SPSS, which is available on the university computers. However, there are many free alternatives to SPSS. If you want to use your own computer, you might consider trying one of these:

- PSPP, which is designed to be nearly identical to SPSS with all the same basic functionality: <https://www.gnu.org/software/pspp/pspp.html>
- JASP, which is more modern, looks nicer and is very easy to use – but less similar to SPSS: <https://jasp-stats.org/>

1 The Theory Crisis

1.1 Weekly Readings

- Muthukrishna, M., & Henrich, J. (2019). A problem in theory. *Nature Human Behaviour*, 3(3), 221–229. <https://doi.org/10.1038/s41562-018-0522-1>
- Eronen, M. I., & Bringmann, L. F. (2021). The Theory Crisis in Psychology: How to Move Forward. *Perspectives on Psychological Science*, 16(4), 779–788. <https://doi.org/10.1177/1745691620970586>

1.2 Tutorial

1.2.1 Find a Group

Pick a card that matches your study program:

- Psychology
- Sociology
- Organizational Studies
- Global Management of Social Issues
- Human Resource Studies



Pick a card that matches your *preferred or default* **role in team work** (you don't have to stick to this role):

- Leader / Visionary
- Monitor / Arbitrator
- Facilitator / Notetaker / Timekeeper
- Devil's Advocate



Now, mingle and try to find group mates with *similar* study programs, and complementary roles in team work. Groups should comprise 3-4 people.

1.2.2 Find a topic

With your group, prepare a text document. At the top of the first page, write:

- Tutorial number (Tutorial 1)
- Group name (come up with your own name)
- Your names and student numbers
- Date

The purpose of this assignment is to find a topic of interest that you want to theorize about, identify what you want your eventual theory to explain, and collect some literature.

1.2.2.1 Step 1

Credit: This tutorial is based on Noah van Dongen's course "Fundamentals: Becoming better at explaining and understanding".

As a group, choose a topic to theorize about. Choose a topic within your field of research, or one that touches upon your field of research. The topic may be quite general, such as “the psychopathology of depression” or “territorial behavior”. It could also be more specific, such as “helping behavior in humans” or “empathy in conflict resolution”. The two restrictions are that A) it can be related to something observable, directly or indirectly. Specifically, you should be able to learn something about the topic you have chosen through observations or experiments. B) You should be able to find sufficient literature (e.g., research) on the topic (see the Fourth part of this assignment). Furthermore, it is recommended to have some experience or knowledge about the topic you choose. Note that you don’t have to be a psychology student to choose (some aspect of) psychology, but it makes things easier if you are at least familiar with some of the research that has been done on the topic. Also, it helps if you choose a topic you find interesting and don’t mind thinking about. Write down the topic you have chosen and explain in a few sentences (max 100 words) what the topic is about.

1.2.2.2 Step 2

Think of three **phenomena** (Nak et al., 2026) related to your topic you would like to explain. Describe these phenomena in a few sentences. List the phenomena’s components/characteristics. Indicate on what the material is based (e.g., a published empirical study, Wikipedia, personal experience, intuition). Note that you do not need to explain these phenomena yet. These phenomena should be as independent of theory (any theory) as possible. It could be that scientists disagree on the explanation of the phenomena, but this should not mean that they also disagree on the existence of the phenomena.

1.2.2.3 Step 3

Do a basic literature search on your chosen topic. Make sure that you have both general sources (e.g., textbook chapters and encyclopedia entries, such as Wikipedia) and specific sources (e.g., scientific journal articles). The aim is to have five highly relevant sources. If you cannot find five sources, check with your groupmates and the tutorial teacher and consider picking a different topic.

List the sources you have found and add hyperlinks or other reference information. For each source, make a note of why you think it is relevant. A few words or a sentence is enough. You do not have to read the whole articles yet.

Having a good search strategy is important. Indicate which search engines and/or databases you used (e.g, Google, Google Scholar, Web of Science) and the (combinations of) search terms you used.

1.2.3 Example

Tutorial 1

Group: The Tacky Theoreticians

Names: Caspar van Lissa (123456), Noah van Dongen (789012)

Date: 02-08-2026

1.2.3.1 Step 1

Social behavior of humans. Homo sapiens are social animals that respond to one another and their behavior is shaped by others that are in their direct vicinity. People also respond differently to situations when they are alone versus when there are other people present. The development of a person's (cognitive) behavior is also in part determined by the presence and behavior of other people. I think my main focus will be on helping behavior.

1.2.3.2 Step 2

- A. Bystander effect. The probability that somebody helps a person in need decreases with the number of other people present (Blair, Thomson, & Wuensch, 2005). The individual components/characteristics are:
 - 1. Without other people present, the tendency (probability or frequency) that a person helps a victim is high.
 - 2. The tendency to help a victim decreases monotonically with the number of other people (bystanders) present.
 - 3. The tendency to help a victim never drops to 0.
- B. Bystander effect when the victim is in danger. The bystander effect varies across different types of situations. The bystander effect is attenuated when the situation is perceived as dangerous for the victim (Fischer et al., 2011). The individual components/characteristics are:
 - 1. Without other people present, the tendency (probability or frequency) that a person helps a victim is high.
 - 2. The tendency to help a victim decreases monotonically with the number of other people (bystanders) present.
 - 3. The tendency to help a victim never drops to 0.
 - 4. The size of the bystander effect is decreased when the situation is perceived as dangerous to the person in need of help.
- C. Bystander effect and danger to the helper. The probability of helping a victim declines as the dangerousness of helping increases (e.g., violent crime with a weapon), in comparison to situations that are less dangerous for the person offering help (e.g., sexual assault with no weapon) (Hart & Miethe, 2008). The individual components/characteristics are:

1. Without other people present, the tendency (probability or frequency) that a person helps a victim is high.
2. The tendency to help a victim decreases monotonically with the number of other people (bystanders) present.
3. The tendency to help a victim never drops to 0.
4. The size of the bystander effect is decreased when the situation is perceived as dangerous to the person in need of help.
5. The size of the bystander effect varies with situational context. The bystander effect is smaller for sexual assault without a weapon than for violent crime with a weapon.

1.2.3.3 Step 3

Wikipedia page: https://en.wikipedia.org/wiki/Bystander_effect

- a. An encyclopedia page that contains general information and lists additional references Britannica page: <https://www.britannica.com/topic/bystander-effect/Diffusion-of-responsibility>
- b. An encyclopedia page that contains general information Fischer, P., Krueger, J. I., Greitemeyer, T., Vogrincic, C., Kastenmüller, A., Frey, D., Heene, M., Wicher, M., & Kainbacher, M. (2011). The bystander-effect: A meta-analytic review on bystander intervention in dangerous and non-dangerous emergencies. *Psychological Bulletin*, 137(4), 517–537. <https://doi.org/10.1037/a0023304>
- c. Is a meta-analysis of studies on the bystander effect and thus contains many references to studies on the bystander effect.

We used Google, Google Scholar, and Web of Science to search for relevant literature. We first used the search term “bystander effect” on Google Scholar, which already produced many hits (39100). We looked at the first three pages and selected those with titles that indicated they were about the bystander effect in (social) psychology. The expectation was that literature on the bystander effect would also be related to social behavior in a more general sense. We then searched on Web of Science to see if any further interesting titles appeared on the first three pages that we missed in Google Scholar. Finally, we searched with Google using the same search term to find more general non-scientific literature that could be of interest, again looking at titles on the first three pages. We think this was an adequate strategy, because we were only searching for a few information-dense and general items and not an exhaustive literature search. We expected that these would show up on the first few pages of the searches, because they most likely would have the term “bystander effect” in the title (Google, Google Scholar, Web of Science) and many places throughout the text (Google). Based on the name of the journal, we selected those that were about (social) psychology.

References

- Blair, C. A., Thompson, L. F., & Wuensch, K. L. (2005). Electronic Helping Behavior: The Virtual Presence of Others Makes a Difference. *Basic and Applied Social Psychology*, 27(2), 171–178. https://doi.org/10.1207/s15324834basp2702_8
- Fischer, P., Krueger, J. I., Greitemeyer, T., Vogrincic, C., Kastenmüller, A., Frey, D., Heene, M., Wicher, M., &
- Kainbacher, M. (2011). The bystander-effect: A meta-analytic review on bystander intervention in dangerous and non-dangerous emergencies. *Psychological Bulletin*, 137(4), 517–537. <https://doi.org/10.1037/a0023304>
- Hart, T. C., & Miethe, T. D. (2008). Exploring bystander presence and intervention in nonfatal violent victimization: When does helping really help? *Violence and Victims*, 23(5), 637–651. <https://doi.org/10.1891/0886-6708.23.5.637>

1.3 Homework

This week, you will select one “canonical version” of an existing theory, or draft your own theory from scratch (optional).



Roll Your Own Theory

The assignment assumes that you will select an existing theory to improve upon. If you draft your own instead, go through all the steps *as if* you are starting from an existing theory - but instead of selecting sources from a published theoretical paper, you write your own. You may find it helpful to:

- Select relevant quotes from *empirical papers* instead, to substantiate the assumptions of your theory.
- Compare your work to that of other groups who have started with an existing theory.
- Come back and revise the work you’ve done this week, once you have gained a better understanding of the next steps in coming weeks.

Credit: This assignment is based on the Proposition Based Theory Specification instructions (Glöckner et al., 2025).

Read pages 1-4 of the [Proposition Based Theory Specification \(PBTS\) Guideline](#).

1.3.1 PBTS Step 1

With your group, select source material for the theory (and specific version of it) to be specified.

- a. Decide which theory, and which version of the theory (e.g., original, revised, or latest) will be specified, including the name of the theory, the author, and the original reference. For example, I used Self-Determination Theory, a leading theory of motivation in social psychology.
- b. Select one canonical source for the theory specification (e.g., the original article or book chapter). If the theory is not fully specified in one source, or if you are drafting your own theory from scratch, select 2-3 key sources instead, such as relevant journal papers, or textbook chapters. For example, I used Ryan (2023).
- c. If the original theory is highly extensive (e.g., published as a book), simplify the task by focusing on its core content as represented in typical journal articles or summary chapters.
- d. Select specific lines within the source material that cover the theory's key causal relations, concept definitions, operationalizations, and auxiliary assumptions. Create a spreadsheet [like this](#) with these selected sentences, label them (e.g., S1) and document the paper, page number, and ideally line number. The final version of Table 1 should include all agreed sources with page/line information, DOI, and, for books, the edition used. For example, see [this spreadsheet](#)
- e. All raters should agree on the selected sources. In cases of uncertainty, debate whether the sentence is indeed relevant to the theory.
- f. Write 100 words on your approach to Source Selection for your final paper.

2 What Makes a Good Theory?

2.1 Weekly Readings

- Meehl, P. E. (1990). Appraising and Amending Theories: The Strategy of Lakatosian Defense and Two Principles that Warrant It. *Psychological Inquiry*, 1(2), 108–141. https://doi.org/10.1207/s15327965pli0102_1
- Guest, O. (2024). What Makes a Good Theory, and How Do We Make a Theory Good? *Computational Brain & Behavior*. <https://doi.org/10.1007/s42113-023-00193-2>

2.2 Tutorial

2.2.1 Analyze Theory Using Meehl's Taxonomy

In this exercise, you will evaluate the sentences you selected from a published theory using Meehl's hierarchy of progressively stronger theory specifications. The goal is to further your understanding of the function and level of specificity of each sentence. Meehl's original taxonomy is:

- I. Type of entity postulated
- II. Compositional, developmental, or efficient-causal connections among entities
- III. Signs of first and second derivatives of functional dynamic laws
- IV. Ordering relationships among the derivatives
- V. Signs of mixed second-order partial derivatives (interactions)
- VI. Functional forms (e.g., linear, logarithmic, exponential)
- VII. Trans-situationality of parameters
- VIII. Quantitative relations among parameters
- IX. Numerical values of parameters

To simplify the assignment, we simplify Meehl's taxonomy as Van Lissa et al. (2026) did:

- I. statements about the types of entities postulated (i.e., ontology)
- II. statements about causal connections between those entities
- III. statements about the functional form of those connections (Levels 3–8)
- IV. statements about their specific numerical values (Level 9)

2.2.1.1 Part 1. Classify the theory statements (15 minutes)

First individually, for each selected sentence:

- Read the sentence carefully.
- Decide which of the levels it best represents.

2.2.1.2 Part 2. Evaluate the theory (10 minutes)

As a group:

- Compare your classifications
- Which levels of Meehl's hierarchy are well represented?
- Which levels are missing?
- How strong do you think the theory is according to Meehl's framework?

2.2.1.3 Part 3. Compare theories (15 minutes)

Each group gives one example of every level they identified in their source material, as well as an example of a sentence you were not able to classify (if applicable). Discuss with the class.

2.2.2 Is This Theory “Good”?

Apply Olivia Guest's four-category ontology to critically evaluate a published psychological or social science theory.

2.2.2.1 Part 1. Evaluate Theory along Each Category (10 minutes)

Each group member picks one unique category, and independently answers the guiding questions for that category in Table 1.

Decide whether the theory is “good”, adequate, or “bad” in your category.

2.2.2.2 Part 2. Discuss and Prioritize (15 minutes)

Discuss your findings with your groupmates. Make a score card for the theory, rating it as good, adequate, or bad on the four categories.

Then, prioritize the four categories. How important do you think it is that the theory be “good” on each of the four categories? Use this to guide your further work on the theory during coming weeks.

2.2.2.3 Part 3. Class discussion (15 minutes)

As a class, discuss:

- If you had to choose between a theory that scores highly on Meehl’s hierarchy but poorly on Guest’s ontology, and one that does the opposite, which would you consider the better scientific theory? Why?
- You can pick one criterion out of Meehl’s hierarchy and Guest’s ontology. Which one do you pick, and why?

2.3 Homework

This week, you will extract IF-THEN statements from the sources you selected last week.

Credit: *This assignment is based on the Proposition Based Theory Specification instructions (Glöckner et al., 2025).*

2.3.1 PBTS Step 2

Individually, fill out [Table 2](#). Translate each source sentence into a set of implications (i.e., IF-THEN statements) and all included propositions that together fully describe the theory.

- Propositions are statements concerning measurable and non-measurable concepts that can be true or false, and represent an implication’s antecedence and consequence. E.g., “A person’s needs are satisfied” is a proposition.
 - Propositions can be combined by any logical or mathematical operators, qualifiers, or, if necessary, user-defined uncommon operators (that have to be defined and operationalized).
 - It is often possible to make certain conditions of a theory either part of the propositions or the definition/operationalizations. To keep the propositions simple, these conditions should be part of the definition/operationalizations if they do not belong to the core causal relations of the theory.
 - In this step, remain as close as possible to the original verbal narrative while interpreting the text in accordance with the authors’ intended meaning.
- (f) In the “Reference” column, reference the label of the text snippet from last week’s Table, e.g. S1, S2 ... Sn.
- If the original specification mentions the functional form of a relationship, put this information in the Comment column.

2.3.2 Combining Specifications-BINGO

As a group, read each other's specifications. Select one student's work to be the "starting point" for the definitive version. The author of this version reads each of their implications aloud. Other students put a check mark in front of their implications if these match the implication read by the first student. If they are clearly similar but there are important differences - discuss these and find agreement on which version of the implication to keep in the definitive version.

When the first student is done reading all of their implications, the other students may have some remaining un-checked implications. That is to say: Implications that they coded, but the first student did not. Discuss these to check whether they should be included in the definitive version or not.

The end product should be a complete table of all unique implications that all students (more or less) agree upon.

If there are important disagreements - keep notes, because this is good material for the Discussion section of your final report.



Roll Your Own Theory

If you are making your own theory, do the assignment as described above - but instead of coding selected sentences, each student lists all implications that *they think should be part of the theory*. The group activity then constitutes of discussing these and finding agreement about a set of implications.

3 Strategic Ambiguity

3.1 Weekly Readings

- Frankenhuis, W. E., Panchanathan, K., & Smaldino, P. E. (2023). Strategic Ambiguity in the Social Sciences. *Social Psychological Bulletin*, 18, 1–25. <https://doi.org/10.32872/spb.9923>
- Healy, K. (2017). Fuck nuance. *Sociological Theory*, 35(2), 118-127.

3.2 Tutorial

3.2.1 Ambiguity Battle (30 min, 2x 15)

Collaborate with 4 students, divided into two duos: Theorists VS Challengers. One duo chooses one of the following phenomena and writes down a one-sentence theory to explain the phenomenon:

- Venus effect: most people may hold beliefs that are inconsistent with observable phenomena
- Social facilitation: people do better on tasks when they are with other people rather than when they are doing the task alone.
- The frequency illusion: people notice a specific concept, word, or product more frequently after recently becoming aware of it.
- Barnum effect: individuals give high accuracy ratings to descriptions of their personality that supposedly are tailored specifically to them, yet which are in fact vague and general enough to apply to a broad range of people. This effect can provide a partial explanation for the widespread acceptance of astrology, fortune telling, aura reading, and the Myers-Briggs “personality test”.
- The bystander effect: individuals are less likely to offer help to a victim in the presence of other people.
- Dunning-Kruger effect: people with low ability in a specific area systematically give overly positive assessments of their ability, while high performers tend to underestimate their skills.
- The Group Project Paradox: Everyone in the group feels that they did disproportionately more work than the others.

The other duo brainstorms on “challenges” to the theory, for example:

- Can it explain a null result?
- Can your theory explain cross-cultural variation?
- Can it explain developmental change?
- Can it explain contradictory results in experimental versus observational/naturalistic settings?

You may also describe a specific challenge, for example “My study found that people with high agreeableness do not display the Dunning-Kruger effect”.

After each challenge, the first group may revise their theory - but without violating the constraint that it must be a single sentence.

The theorists win if the challengers cannot come up with new challenges. The challengers win if the theorists cannot amend their theory while respecting the one-sentence constraint.

Switch sides after 10 minutes.

3.2.2 Abstraction VS Nuance (30 min)

Starting with the one-sentence theories from the previous exercise, go around in a circle.

1. In turn, each student mentions their personal experience in relation to the theory (so: 3 or 4 rounds per theory).
2. The other students vote on whether this is consistent with the theory, or whether the theory should be made “more nuanced” to accommodate it.
3. All together, amend the theory if necessary.

Do this for 10 minutes, so for 2-3 theories. Then discuss:

- When did each theory stop becoming better?
- Did any additions genuinely improved explanation?
- Which merely increased complexity?

3.2.3 Ambiguous Causal Claims (30 min)

Causal claims are an important part of (social scientific) theories. However, there are strong norms against explicit discussion of causality in the social science literature. The result is that causal language is often ambiguous. Norouzi et al. (2025) showed that it is possible to automatically extract causal sentences from published papers despite this ambiguity.

The sentences below are examples from the research by Norouzi and colleagues.

With your group, decide for each sentence and keep notes on:

- a. Whether it is Causal (C) or Non-causal (NC)
 - b. Whether it is Ambiguous (Am) or Straightforward (S)
 - c. Whether it is Nuanced (N) or Abstract (Ab)
- We find differences in the behavior of our experimental participants across the three countries, and we find that Hofstede's dimensions help to explain these differences.
 - In the standard symmetric independent private values model in which $v_i(d_i) = 0$, Myerson's (1981) revenue equivalence result shows that, given that the charity never keeps the prize, it maximizes its revenue if it always allocates the prize to the donor with the highest marginal revenue.
 - First, models of information signaling have shown that sequential giving can have beneficial effects for voluntary public good provision when the common value of the public good is uncertain.
 - Even though the level of regulation a facility faces does not depend on the overall size of a facility, the two factors are highly correlated.
 - For example, in social dilemmas with random noise, people lack the ability to judge with certainty if others intended to cooperate or defect (Klapwijk & Van Lange, 2009; Van Lange, Ouwerkerk, & Tazelaar, 2002).
 - The presence of a threshold constraining participants' activities will lead participants to be less holistic in their cognitive orientations than when a threshold is not present.
 - This result is consistent with a previous study which suggests that behaving dishonestly leads to the forgetting of rules.
 - Violent media may cause viewers to perceive greater anger and less gratitude during a cooperative task and as such be less inclined to cooperate.

Solution

- Violent media may cause viewers to perceive greater anger and less gratitude during a cooperative task and as such be less inclined to cooperate. C, S, Ab
- This result is consistent with a previous study which suggests that behaving dishonestly leads to the forgetting of rules. C, S, Ab
- Even though the level of regulation a facility faces does not depend on the overall size of a facility, the two factors are highly correlated. NC, S, Ab
- We find differences in the behavior of our experimental participants across the three countries, and we find that Hofstede's dimensions help to explain these differences. NC, S, N
- The presence of a threshold constraining participants' activities will lead participants to be less holistic in their cognitive orientations than when a threshold is not present. C, Am, N
- First, models of information signaling have shown that sequential giving can have beneficial effects for voluntary public good provision when the common value of the public good is uncertain. C, Am, N
- In the standard symmetric independent private values model in which $v_i(d_i) = 0$, Myerson's (1981) revenue equivalence result shows that, given that the charity never keeps the prize, it maximizes its revenue if it always allocates the prize to the donor with the highest marginal revenue. C, S, N
- For example, in social dilemmas with random noise, people lack the ability to judge with certainty if others intended to cooperate or defect (Klapwijk & Van Lange, 2009; Van Lange, Ouwerkerk, & Tazelaar, 2002). NC, S, N

Discuss:

- How would you go about writing about causality in your work, after having seen these examples?
- What do you think is the difference between ambiguity and nuance?

3.3 Homework

This week, you are going to translate your IF-THEN specifications from previous weeks into a graph - the technical name is a **Directed Acyclic Graph (DAG)**. In this graph, you connect concepts (propositions, in the language of PBTS) with one-sided arrows to indicate the direction of causality.

Note that IF-THEN connections imply causal connections, so the statement "IF a person's need for autonomy is satisfied THEN they experience well-being" implies a causal link of the form {autonomy satisfied} -> {well-being}.

3.3.1 Example: Self-Determination Theory

Let's examine a snippet from self-determination theory, and code it for implied causality. The causes are coded in green, and outcomes in blue:

“For these natural, active **processes of intrinsic motivation and integration** to operate effectively toward **healthy development and psychological well-being**, human beings need particular **nutriments** – both biological and psychological (Ryan, 1995). In the relative absence of such **nutriments**, these natural **processes** will be impaired, resulting in **experiences, development, and behaviors** that are less than optimal” (Lange et al., 2012, p. 417)

The text is a bit ambiguous, and possibly redundant. We see that **processes**, in the first sentence, refers to intrinsic motivation and integration. This invites the question of the relationship between intrinsic motivation and integration - do they go hand in hand (i.e., just two examples of processes set in motion by nutriment)? Are they distinct but similarly related to other constructs? It's not clear. Furthermore, in the second sentence, we find the word **processes** again, but this time without explicit reference to intrinsic motivation and integration. For now, we will assume that intrinsic motivation and integration are distinct but have similar relationships to other constructs, and that the word is used consistently across both sentences.

Nutriments is defined elsewhere in the text - it appears to refer to the three basic needs, as well as biological necessities.

Experiences, development, and behaviors is not well-defined, but we might assume it refers back to **healthy development and psychological well-being**.

In Week 2, we learned to code snippets like this for IF-THEN connections according to PBTS. This would give the following codings:

```
Attaching package: 'theorytools'
```

```
The following objects are masked from 'package:webexercises':
```

```
add_to_pkgdown, use_webex_vignette
```

N	IF	THEN
1	nutriments are absent	intrinsic motivation AND integration are impaired
2	[intrinsic motivation OR integration] is present	healthy development AND psychological well-being
3	needs are satisfied or thwarted	psychological well-being of all people is affected
4	rewards given	intrinsic motivation can decrease
5	external event is expected to thwart the basic needs	external perceived locus of causality
6	external perceived locus of causality	undermined intrinsic motivation
7	external event is expected to support the basic needs	internal perceived locus of causality
8	internal perceived locus of causality	enhanced intrinsic motivation

We can translate these IF-THEN statements into causal links between specific constructs.

This translation reveals some ambiguities. For example, each IF-THEN relationship is often repeated twice; once phrased in the positive and once in the negative. Many propositions are repeated, once phrased in a positive way, and once phrased in a negative way. For example, *human beings [need] nutriments [...] for these processes to operate* positively phrases a link from nutriments to processes, whereas *In the [absence of] nutriments [...] processes will be impaired* negatively phrases the same link. This can be confusing, especially if the two phrasings differ slightly. Here, it seems safe to assume that both sentences have the same meaning.

Another ambiguity is the lack of clear and consistent terminology. For example, in proposition #4, we see “rewards” are related to intrinsic motivation. However, in the remainder of the text, rewards are mentioned as just one example of “external events”. We assumed that “rewards” is used as an example of the broader class of external events (*pars pro toto*).

Another ambiguity: we read that “If [an external event] were expected to thwart [basic needs], it [prompts] an external perceived locus of causality”. Since the theory otherwise does not explain how people might “expect” that external events will affect them, and this seems to be more of a question for cognitive than social psychology - we made a simplifying assumption: external events affect needs, and also perceived locus of causality.

We applied two transformations:

1. Replace the original text with concept labels, e.g. “nutriments are present” and “needs are satisfied or thwarted” are both replaced with “needs”.
2. After this simplification, remove redundant links

Table 3.1: Unique causal connections

from	to
needs	intrinsic_motivation
needs	integration
intrinsic_motivation	healthy_development
intrinsic_motivation	wellbeing

from	to
integration	healthy_development
integration	wellbeing
needs	wellbeing
external_event	intrinsic_motivation
external_event	needs
external_event	locus_of_causality
locus_of_causality	intrinsic_motivation

We translated this table to a DAG:

We can plot the DAG as follows:

```
plot(SDT)
```

Plot coordinates for graph not supplied! Generating coordinates, see ?coordinates for how to

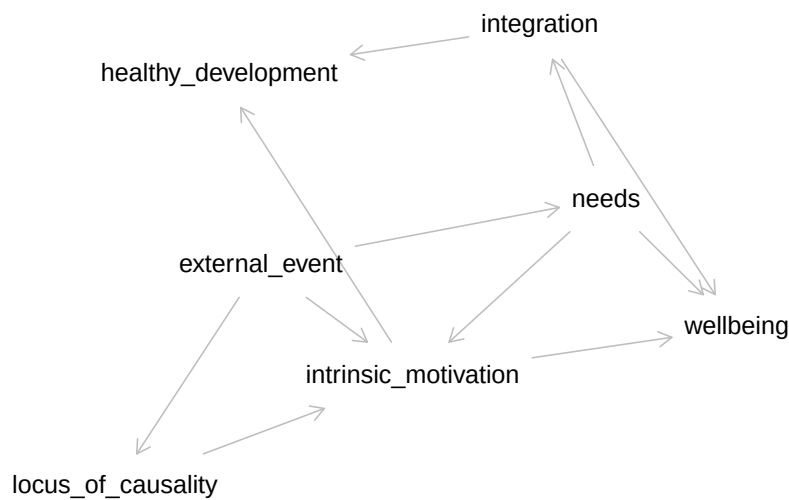


Figure 3.1: Causal diagram implied by SDT

3.3.2 Do It Yourself

Following the example above, you will perform three tasks:

- ☐ Grouping plain-language propositions into a limited set of concepts
- ☐ Revising your **Step_2.csv** table so that it only uses that limited set of concepts
- ☐ Constructing a DAG in R using your revised table.

3.3.2.1 Group Propositions into Concepts

Make a copy of your `Step_2.csv` table, and call it `DAG.csv`.

With your groupmates, discuss:

- Which of the propositions (i.e., the text in the IF- and THEN-columns) are really the same? E.g.: Are “autonomy satisfaction” and “people feel more independent” really the same?
- Are any of them really the same, but positively vs negatively phrased (e.g., “autonomy satisfaction” and “frustration of the need for autonomy”)?
- How would you name them if you merged identical propositions? E.g., you could use `autonomy_satisfaction`, or you could go even more abstract and use `need_satisfaction`, in which case you would use the same label for “autonomy satisfaction”, “competence satisfaction”, and “relatedness satisfaction”.

Obviously, these examples are based on SDT; use your own theory for this assignment.

Note: For technical reasons, it is preferable to use short labels without any spaces or special characters. So use `need_satisfaction`, not `satisfaction` (of different needs, like autonomy).

3.3.2.2 Revise your Table

Now go through your table, `DAG.csv`, and replace every original text with the labels you decided on in the previous step.

3.3.2.3 Make DAG in R

- ☐ Open the folder `patterns_tutorials` you created when following the setup tutorial, `?@sec-setup`.
- ☐ Copy your file `DAG.csv` into this folder.
- ☐ Load the RStudio project file, `patterns_tutorials.Rproj`.
- ☐ In the bottom right panel of RStudio, click the `Files` tab, and open the file `make_dag.R`.

Run the following code line by line:

```
library(theorytools)
library(dagitty)
my_theory <- read.csv("DAG.csv", stringsAsFactors = FALSE)[, c(2:3)]
names(my_theory) <- c("from", "to")
# Remove redundant statements
my_theory <- my_theory[!duplicated(my_theory), ]
```

```
my_theory <- dagitty(  
  paste0("dag {",  
    paste0(my_theory$from, " -> ", my_theory$to, collapse = "\n"),  
    "}")  
)  
plot(my_theory)
```

4 Latent Theory in Published Papers

4.1 Weekly Readings

- Norouzi, R., Kleinberg, B., Vermunt, J. K., & Lissa, C. J. van. (2025). Capturing causal claims: A fine-tuned text mining model for extracting causal sentences from social science papers. *Research Synthesis Methods*, 16(1), 139–156. <https://doi.org/10.1017/rsm.2024.13>
- Norouzi, R., Kleinberg, B., Vermunt, J., & Van Lissa, C. J. (2026). Can we Automate Social Science Concept Harmonization? A Fine-Grained Review of Performance and Failure Modes (G9pax_v2). *PsyArXiv*. https://osf.io/preprints/psyarxiv/g9pax_v2/

4.2 Tutorial

[TO DO: Rasoul]

4.3 Homework

Take the graph you produced during the tutorial, and compare it to the DAG you made in `?@sec-homeworkdag`.

Are there any concepts or connections in this new graph that your DAG is missing? Do you want to add these? If so, why (not)? Keep notes on your decision making process for your final paper.

5 Formal Theory Specification

5.1 Weekly Readings

New names:

```
* `` -> `...15`  
* `` -> `...16`  
* `` -> `...17`  
* `` -> `...18`  
* `` -> `...19`  
* `` -> `...20`  
* `` -> `...21`
```

- Muthukrishna, M., & Henrich, J. (2019). A problem in theory. *Nature Human Behaviour*, 3(3), 221–229. <https://doi.org/10.1038/s41562-018-0522-1>
- Eronen, M. I., & Bringmann, L. F. (2021). The Theory Crisis in Psychology: How to Move Forward. *Perspectives on Psychological Science*, 16(4), 779–788. <https://doi.org/10.1177/1745691620970586>

5.2 Tutorial

In this tutorial, you will go through some of the steps of theory formalization, following the Abductive Formal Theory Construction (AFTC) framework. We will use Self-Determination Theory (SDT) as a guiding example.

The code chunks in this tutorial are also available in the `patterns_principles_exercises` project, in the file `tutorial_formal_theory.R`.

5.2.1 Stage 1: Generating Theory

5.2.1.1 Establish the Phenomena

Below are two excerpts from Ryan (2023):

“Deci’s early studies (e.g., 1971, 1972) were provocative in showing that contingent extrinsic rewards could have negative effects on subsequent interest and persistence (see Ryan, Ryan, & Di Domenico, 2019).”

“These “undermining” effects of rewards on intrinsic motivation did not occur invariantly but were conditional. As summarized in Deci and Ryan (1980a), rewards have negative effects on intrinsic motivation when used in controlling ways, that is, when applied in order to externally pressure or induce people to perform certain behaviors or meet certain standards. Such controlling reward contingencies can undermine the experience of autonomy essential to intrinsic motivation. In contrast, rewards can have more positive, or at least no negative, effects on intrinsic motivation when experienced as informational”

Discuss with your group, and describe concisely in your own words, what “phenomena” this excerpt describes.

5.2.1.2 Generate an Initial Verbal Theory

You have previously worked with a DAG representing SDT. The vignette [Formalizing Self-Determination Theory](#) describes how this FAIR theory was created based on the original Self-Determination Theory (**deciSelfDeterminationTheory2012?**). It was specified as a Directed Acyclic Graph, or DAG: a diagram that specifies the causal relations proposed by a theory. Such diagrams can be used to derive hypotheses, select control variables for causal inference, and conduct simulation studies (as we will do in this tutorial).

```
library(theorytools)
```

Attaching package: 'theorytools'

The following objects are masked from 'package:webexercises':

```
add_to_pkgdown, use_webex_vignette
```

```
library(dagitty)
sdt <- dagitty("
dag {
  external_event
  healthy_development
  integration
  intrinsic_motivation
  locus_of_causality
```

```

needs
wellbeing
external_event -> intrinsic_motivation
external_event -> locus_of_causality
external_event -> needs
integration -> healthy_development
integration -> wellbeing
intrinsic_motivation -> healthy_development
intrinsic_motivation -> wellbeing
locus_of_causality -> intrinsic_motivation
needs -> integration
needs -> intrinsic_motivation
needs -> wellbeing
}"))
sdt

```

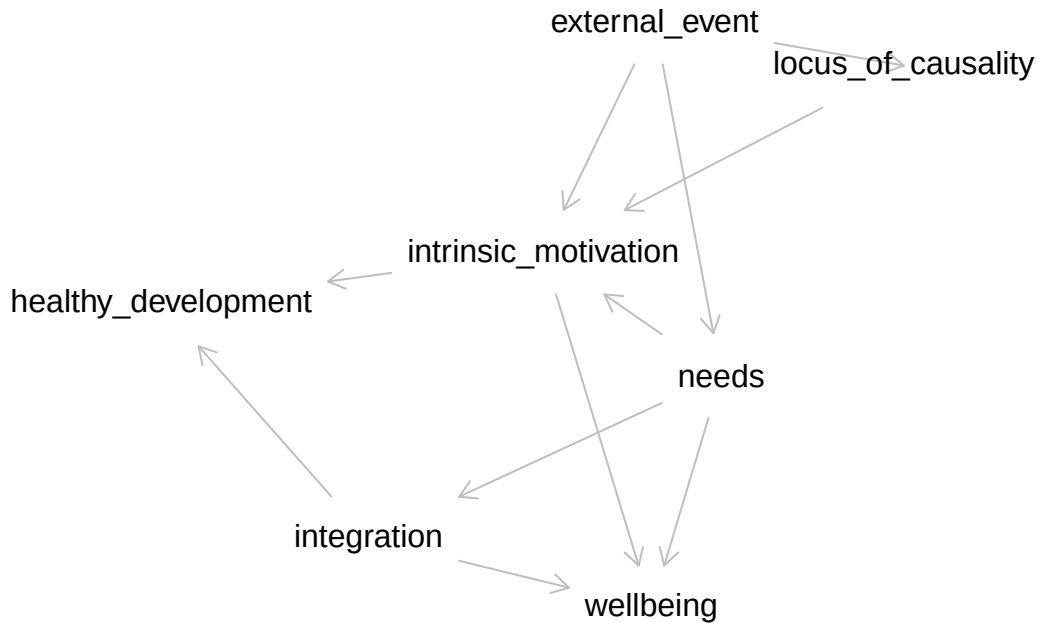
```

dag {
external_event
healthy_development
integration
intrinsic_motivation
locus_of_causality
needs
wellbeing
external_event -> intrinsic_motivation
external_event -> locus_of_causality
external_event -> needs
integration -> healthy_development
integration -> wellbeing
intrinsic_motivation -> healthy_development
intrinsic_motivation -> wellbeing
locus_of_causality -> intrinsic_motivation
needs -> integration
needs -> intrinsic_motivation
needs -> wellbeing
}

```

We can also plot this DAG:

```
plot(sdt)
```



Discuss with your group:

- Which portions of the DAG are relevant to the phenomenon you've identified?
 - Make a simplified version of the DAG that leaves out all irrelevant components.
- Is the simplified DAG sufficient to explain your identified phenomenon? If it helps, think of the arrows as IF/THEN links. If the causes are present, does the outcome always follow?
 - If not: What additions would you need to make to the DAG to properly describe the phenomenon?

5.2.1.3 Formalize the Initial Theory

We will construct a very basic formalization for the section of SDT that you have selected.

First, we will simplify the DAG. You can do this by hand; suppose we are interested in studying the effect of intrinsic motivation on well-being. We do not need the entire theory to hypothesize about this specific relationship. We can simplify the DAG to derive a restricted version of the theory that includes all variables that might confound the relationship of interest between intrinsic motivation and well-being:

```
sdt_pruned <- dagitty("
dag {
  intrinsic_motivation -> wellbeing
  external_event -> intrinsic_motivation
  needs -> intrinsic_motivation
  needs -> wellbeing
}")
sdt_pruned
```

```
dag {
  external_event
  intrinsic_motivation
  needs
  wellbeing
  external_event -> intrinsic_motivation
  intrinsic_motivation -> wellbeing
  needs -> intrinsic_motivation
  needs -> wellbeing
}
```

Do the same for the phenomenon you identified. When you have done so, we can now create some code to generate a synthetic dataset based on this simplified DAG, using the `simulate_data()` function. This function samples random values for exogenous variables and computes endogenous variables as functions of their predictors (linear functions, by default). By default, regression coefficients are randomly sampled in the range $[-0.6, +0.6]$ unless specific values are provided by the user. You might improve the model by choosing more realistic parameter values, for example, using the conventional values for null (0), small (.2), or medium (.4) effect sizes. By default, normal distributions are assumed for exogenous variables and residual error, unless other distributions are specified.

Here, we first generate the code, then examine it in the console, then write it to a file, and finally, run that file to create the simulated data.

```
sim_code <- simulate_data(sdt_pruned, n = 100, run = FALSE)
sim_code
```

```
[1] "# Set random seed"
[2] "set.seed(1131115390)"
[3] "# Set simulation parameters"
[4] "n <- 100"
[5] "# Simulate exogenous nodes"
```

```

[6] "external_event <- rnorm(n = n)"
[7] "needs <- rnorm(n = n)"
[8] "# Simulate endogenous nodes"
[9] "intrinsic_motivation <- 0.15 * needs + 0.32 * external_event + rnorm(n = n)"
[10] "wellbeing <- 0.54 * needs - 0.25 * intrinsic_motivation + rnorm(n = n)"
[11] "df <- data.frame("
[12] "external_event = external_event,"
[13] "intrinsic_motivation = intrinsic_motivation,"
[14] "needs = needs,"
[15] "wellbeing = wellbeing"
[16] ")"
```

```

writeLines(sim_code, "formal_model.R")
source("formal_model.R")
```

Put this code into a separate R-file. When you run it all, it will create an object called `df`, which is a table with 100 observations for all variables. Each exogenous variable (causes) is generated by drawing random samples from a normal distribution. Each endogenous variable (effects) is generated by multiplying those exogenous variables with a regression coefficient; larger values mean that the exogenous variable has a larger effect. Then, random noise is added, which represents variability due to other causes not part of your model. If you want this random variability to be smaller, change the `sd` argument of the `rnorm()` function.

5.2.1.4 Evaluating the Initial Theory

Examine the properties of the dataset using any of the following methods, or some of your own choosing:

5.2.1.4.1 Examining descriptive statistics

You can examine the distributions of individual variables using descriptive statistics.

Do not try all of these, instead, discuss which one should capture the pattern you're looking for in the data:

- Univariate distribution, or patterns related to the center and spread of the distribution of one variable: Descriptive statistics and density plots
- Relationships between two variables: Scatter plots and bivariate regression
- Interaction effects (the effect of one variable depends on the value of another variable): Regression with an interaction term (difficult to plot, because that would be a 3-dimensional plot)

5.2.1.4.2 Univariate Distribution

For example, to obtain a summary of all variables in the dataset:

```
worcs::descriptives(df)
```

	name	type	n	missing	unique	mean	median
1	external_event	numeric	100	0	100	-0.07419535	-0.1741368
2	intrinsic_motivation	numeric	100	0	100	-0.15860162	-0.2173373
3	needs	numeric	100	0	100	0.09889495	0.1152681
4	wellbeing	numeric	100	0	100	0.28180240	0.1904639

	mode	mode_value	sd	v	min	max	range	skew
1	-0.1741368	NA	0.9688439	NA	-2.219908	1.944550	4.164457	0.071086266
2	-0.2173373	NA	1.1204311	NA	-2.695574	2.965118	5.660692	0.189862670
3	0.1152681	NA	0.9420906	NA	-2.548268	2.513896	5.062164	-0.005531599
4	0.1904639	NA	1.0957977	NA	-2.251550	3.541369	5.792919	0.280579240

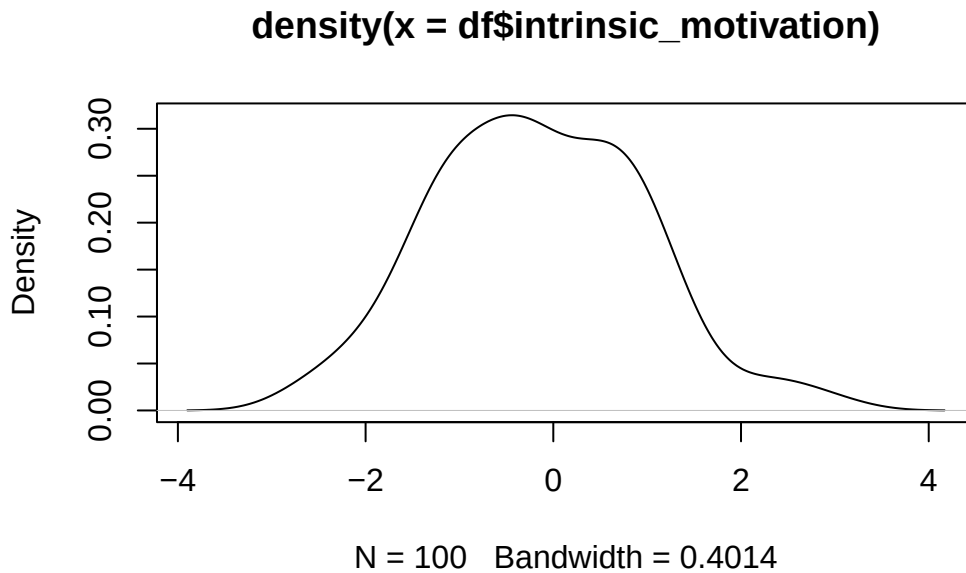
	skew_2se	kurt	kurt_2se
1	0.14724984	-0.68043082	-0.71125500
2	0.39328620	-0.17778337	-0.18583713
3	-0.01145829	0.06852363	0.07162782
4	0.58119873	0.34543934	0.36108808

This gives you information such as the **minimum**, **maximum**, **median**, **mean**.

Use these statistics to look for interesting properties of the data. For example: Does a variable have a particularly high or low mean? Is there a lot of variation between observations? Are some categories much more common than others?

You can also use **base R plots** to visually examine the univariate distribution of an individual variable. For a continuous variable, you can examine its distribution with a density plot:

```
plot(density(df$intrinsic_motivation))
```



Use the plots to look for interesting patterns. For example: Is a distribution symmetric or skewed? Are there possible outliers?

5.2.1.4.3 Relationship between Two Variables

You can test the effect of one variable on another by running a bivariate regression, e.g.:

```
summary(lm(intrinsic_motivation~external_event, df))
```

Call:

```
lm(formula = intrinsic_motivation ~ external_event, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.60590	-0.68605	0.01491	0.78650	2.32123

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.1287	0.1058	-1.216	0.227090
external_event	0.4036	0.1095	3.687	0.000373 ***

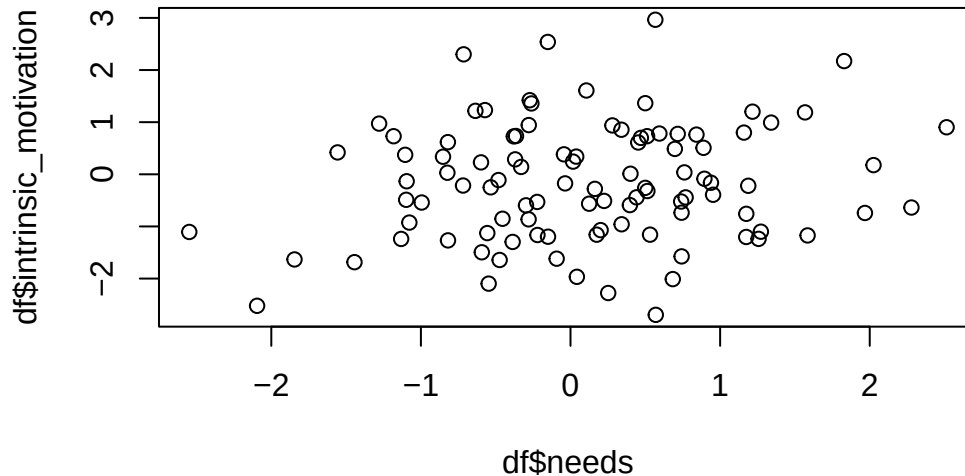
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.055 on 98 degrees of freedom
Multiple R-squared: 0.1218, Adjusted R-squared: 0.1128
F-statistic: 13.59 on 1 and 98 DF, p-value: 0.0003725

If the value of $\Pr(>|t|)$ for `external_event` is smaller than .05, the effect is significant. If the value of `Estimate` is positive, then the effect is positive - if it is negative, then the effect is negative.

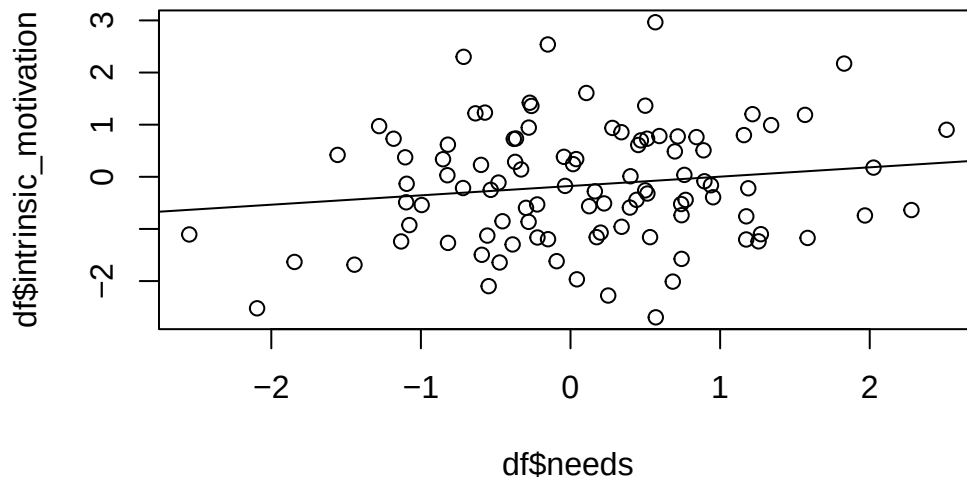
To plot the relationship between two continuous variables, you can use a scatterplot:

```
plot(df$needs, df$intrinsic_motivation)
```



You can add the regression line into this plot to make the direction of the relationship easier to see:

```
plot(df$needs, df$intrinsic_motivation)  
abline(lm(intrinsic_motivation ~ needs, df))
```



5.2.1.4.4 Interaction Effects

You can test whether the effect of one variable on another is contingent on the value of a third variable by running multiple regression regression with an interaction effect, e.g.:

```
summary(lm(intrinsic_motivation~external_event*needs, df))
```

Call:

```
lm(formula = intrinsic_motivation ~ external_event * needs, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.70385	-0.76529	0.00465	0.66256	2.30890

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.14706	0.10577	-1.390	0.167632
external_event	0.40384	0.11299	3.574	0.000552 ***
needs	0.20107	0.11272	1.784	0.077614 .
external_event:needs	0.03139	0.11606	0.270	0.787425

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.049 on 96 degrees of freedom

Multiple R-squared: 0.15, Adjusted R-squared: 0.1235

F-statistic: 5.648 on 3 and 96 DF, p-value: 0.001316

If the value of $\Pr(>|t|)$ for `external_event:needs` is smaller than .05, the interaction effect is significant. That means that the effect of `external_event` on `intrinsic_motivation` depends on the value of `needs`.

5.2.2 Stage 2: Developing Theory

The final step is to make changes to your formal theory, to make it better produce the kind of data patterns that your phenomenon implies.

You will probably not have time to do so, but it is possible to make changes to the file "formal_model.R".

For example, you can:

5.2.2.1 Generating a binary variable

You can generate a binary variable using `rbinom()`. For example, if you want to conceptualize `external_event` as the presence or absence of a reward, you could simulate a variable where approximately 50% of observations have a value of 0 and 50% have a value of 1:

```
external_event <- rbinom(n = n, size = 1, prob = 0.5)
```

5.2.2.2 Generating an interaction effect

To make the effect of `external_event` on `intrinsic_motivation` depend on `needs` (e.g., does this external event satisfy needs?), add an interaction term to the equation that predicts `intrinsic_motivation`:

```
intrinsic_motivation <-  
  0.56 * external_event +  
  0.59 * needs +  
  0.80 * external_event * needs +  
  rnorm(n = n)
```

The 0.80 specifies the strength of the interaction.

In this example, the effect of `external_event` on `intrinsic_motivation` becomes stronger as `needs` increases.

5.2.2.3 Generating a quadratic effect

To create a nonlinear relationship, include a squared version of a variable:

```
intrinsic_motivation <-  
  0.56 * external_event +  
  0.59 * needs -  
  0.70 * needs^2 +  
  rnorm(n = n)
```

The negative coefficient on `needs^2` creates an inverted n-shaped relationship: intrinsic motivation initially increases with needs, but at sufficiently high levels of needs it begins to decrease.

A positive quadratic coefficient instead produces a u-shaped relationship.

5.3 Homework

Apply the steps from the tutorial to the DAG you created for your own theory in `?@sec-makedag`.

Note that you do not have to apply *all* steps. Think of this week's tutorial as a palette from which you can pick the techniques you need for your specific theory. See how far you can get, and critically reflect on the process.

6 Machine Learning-Informed Theory Construction

6.1 Weekly Readings

- Van Lissa, C. J. (2022). Developmental data science: How machine learning can advance theory formation in Developmental Psychology. *Infant and Child Development*, 32(6), 1–12. <https://doi.org/10.1002/icd.2370>
- Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5), Article 5. <https://doi.org/10.1038/s42256-019-0048-x>

6.2 Tutorial

The topic of this week is to use machine learning to detect patterns in a dataset pertaining to your theory of interest. I have prepared a reproducible example for you based on Self-Determination Theory in the `patterns_principles_exercises` project. The purpose of this tutorial is **not** to give you a foundation in machine learning; that is too complex a topic. However, you should be able to run the code, you are allowed to play around with it and try making some changes, and you might be able to adapt and reuse it for future research projects.

It will probably surprise you that preparing the data for analysis is the most difficult part of this tutorial. Expect to spend most of your time on this step - and **make sure to get the tutorial teacher's approval for your cleaned dataset before you go home.**

6.2.1 Step 1: Find a Dataset (30 minutes)

Search through [Google Database Search](#) for a dataset that covers some variables that can be considered operationalizations of the concepts in your theory. One approach is to search the database for the name of your theory. Another approach is to choose one, two, or three construct names from your theory, and search for those.

6.2.1.1 SDT Example: Codebook

I used the search term "self determination theory", and read the descriptions of the datasets until I found the one by Nguyen and Xiem (2026) on “mental health, academic motivation, and perceived academic performance among Vietnamese students in online learning”. It contains operationalizations of a few SDT constructs in the specific context of online learning (thus, we can debate how well this generalizes to SDT in general settings). That’s about as good an example as you can expect to find on short notice. My dataset contains the following variables:

Variable	Description	Type	Coding . . . Range
On-line	Questionnaire completed online	Categorical	0 = No; 1 = Yes
gender	Respondent gender	Categorical	1 = Male; 2 = Female; 3 = Other
year	Current academic year	Ordinal categorical	1 = Year 1; 2 = Year 2; 3 = Year 3; 4 = Year 4; 5 = Other
field	Respondent’s field of study	Nominal categorical	1 = Economics and Business; 2 = Engineering and Technology; 3 = Languages and Linguistics; 4 = Health Sciences; 5 = Education; 6 = Culture and Arts; 7 = Information Technology; 8 = Other
class_period	Average number of online learning class periods per week	Ordinal categorical	1 = 1–3 periods; 2 = 4–6 periods; 3 = 7–9 periods; 4 = 10 or more periods
COM	Personal competence in using the Internet; mean of COM1–COM6	Continuous	1–5; higher = greater Internet competence
REL	Online relatedness; mean of REL1–REL8	Continuous	1–5; higher = greater relatedness

Variable	Description	Type	Coding . . . Range
AUT	Autonomy in Internet use; mean of AUT1–AUT7	Continuous	1–5; higher = greater autonomy
MEH	Mental health/well-being; mean of MEH1–MEH5	Continuous	1–5; higher = better mental health
ACM	Academic motivation for online learning; mean of ACM1–ACM7	Continuous	1–5; higher = greater academic motivation
ACP	Perceived academic performance; mean of ACP1–ACP5	Continuous	1–5; higher = better perceived academic performance

6.2.2 Step 2: Preprocess the Data (30 minutes)

You will find a file called `prepare_nguyen_2026.R` in the `patterns_principles_exercises` project. Open that file. Each section of code performs a specific data preprocessing step, as explained in the comments. Work through the file one step at a time to prepare your own dataset for analysis. Note that, for real research projects, you would probably do a lot more data cleaning than this - including solving problems specific to your dataset. For the present tutorial, forget about any non-standard problems. Just remove any variables that are not well-behaved.

6.2.3 Step 3: Run the Analysis (20 minutes)

You will find a file called `analysis_nguyen_2026.R` in the `patterns_principles_exercises` project. Open that file, and skim the comments to get a sense of the analysis steps. If you have performed the data cleaning correctly, you should be able to run that file in its entirety without errors, and get machine learning results for your own theory.

6.2.4 Step 4: Interpreting the Results (10 minutes)

We used four machine learning models that capture patterns in the data in different ways. Many other machine learning techniques exist, but most of these are common (except symbolic regression, but it's very interesting nonetheless). In this case, we used:

- **LASSO regression**, which models the outcome as a linear function of the predictors (just like normal regression) - but it shrinks the coefficients of less important predictors toward zero. It is easy to interpret because you can read the table of coefficients to evaluate the size and sign of the effect of included predictors, just like a table of regression coefficients. Irrelevant predictors are eliminated from the model.
- A **regression tree**, which captures patterns by repeatedly splitting the data into groups based on predictor values, producing a set of decision rules that can be interpreted directly. Note that, in this tree structure, an interaction effect like $X_1 \times X_2$ can be represented by first splitting on one interacting variable X_1 , and then on the other, X_2 . A nonlinear effect can be represented by splitting on the same variable at different values.
- **Random forests** combine predictions from many regression trees grown on slight permutations of the sample. They also capture potential interactions and nonlinear effects, and often capture much more detail of the patterns in data than a single tree, at the cost of a loss of interpretability. You can explain a forest's predictions using variable importance and partial dependence plots, which describe which predictors matter most for accurate predictions, and how each predictor affects the predictions while averaging across all other predictors, respectively.
- **Symbolic regression** constructs a mathematical expression, given a set of variables and mathematical operators, that reproduces patterns in the data. You can allow it to construct any kind of function, including nonlinear and interacting functions. The final model is an explicit equation, which can be interpreted at face value.

6.2.4.1 Model Fits

The first thing we do is interpret the table with model fits. Here, we use R^2 - which is a bit similar to the R^2 you might know as “explained variance” from linear regression. One key difference is that we have R^2_{train} and R^2_{test} . The R^2_{train} can be straightforwardly interpreted as the percentage of variance explained in the data the model already saw during training. The R^2_{test} is a bit different: it looks at the variance explained by a specific model *in new data* (the test data), relative to just using the mean of the training data to predict new observations.

Model	R^2_{train}	R^2_{test}
LASSO	0.65	0.69
Random Forest	0.76	0.70
Regression Tree	0.63	0.66
Symbolic regression	0.49	0.57

To pick the best model, it's most useful to look at R^2_{test} , because R^2_{train} can be overly optimistic - a sufficiently complex model can literally memorize the correct value for all the training data, achieving an $R^2_{train} = 1.00$ but an abysmal R^2_{test} . The R^2_{test} does not suffer from this bias, and it gives you an impression of how well your model generalizes to new data.

Note that according to this metric, the random forest model is best. However, a random forest model is not interpretable - although you can analyze its predictions using measures like variable importance and partial dependence plots. Note that the LASSO model is almost equally good - and it is very straightforwardly interpretable. From a theory construction perspective, you might thus choose the LASSO model because it combines near-peak performance with high interpretability. Also note that the differences between models are not that big. Even a symbolic regression model - which is just a linear function of one predictor (see below) - performs quite well.

6.2.4.2 Interpret LASSO

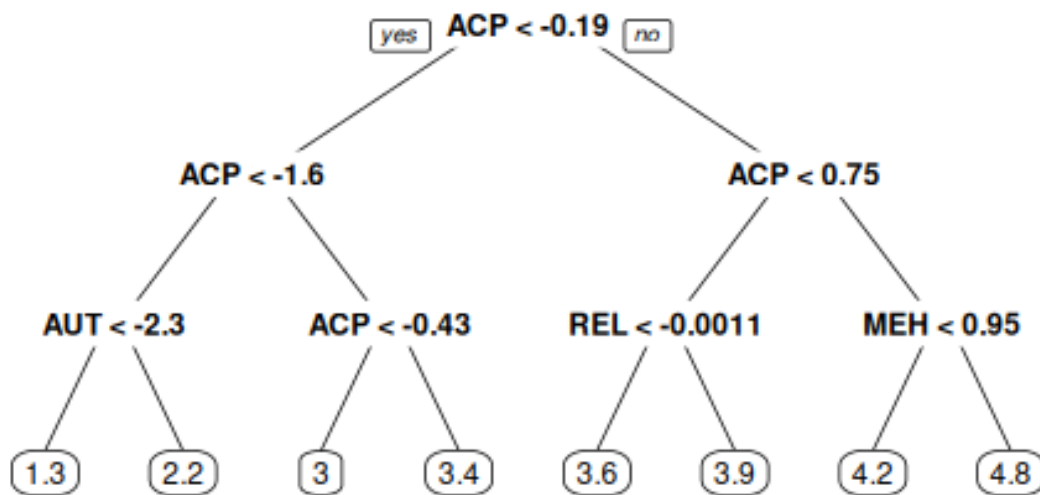
To interpret the LASSO model, we look at its regression coefficients. These are interpreted the same as other regression coefficients, except that these are all biased towards zero:

Predictor	b_{lse}
(Intercept)	3.60
ACP	0.43
MEH	0.12
AUT	0.08
REL	0.05
COM	0.00
Online	0.00
genderM	0.00
genderF	0.00
genderO	0.00
year	0.00
fieldEngineering.and.Technology	0.00
fieldLanguages.and.Linguistics	0.00
fieldHealth.Sciences	0.00
fieldEducation	0.00
fieldCulture.and.Arts	0.00
fieldInformation.Technology	0.00
fieldOther	0.00
class.per.week	0.00

We see that ACP, MEH, AUT, and REL are retained as relevant predictors, of which ACP is by far the most important one.

6.2.4.3 Interpret a Tree

You can interpret a regression tree by plotting it:



The plot tells us:

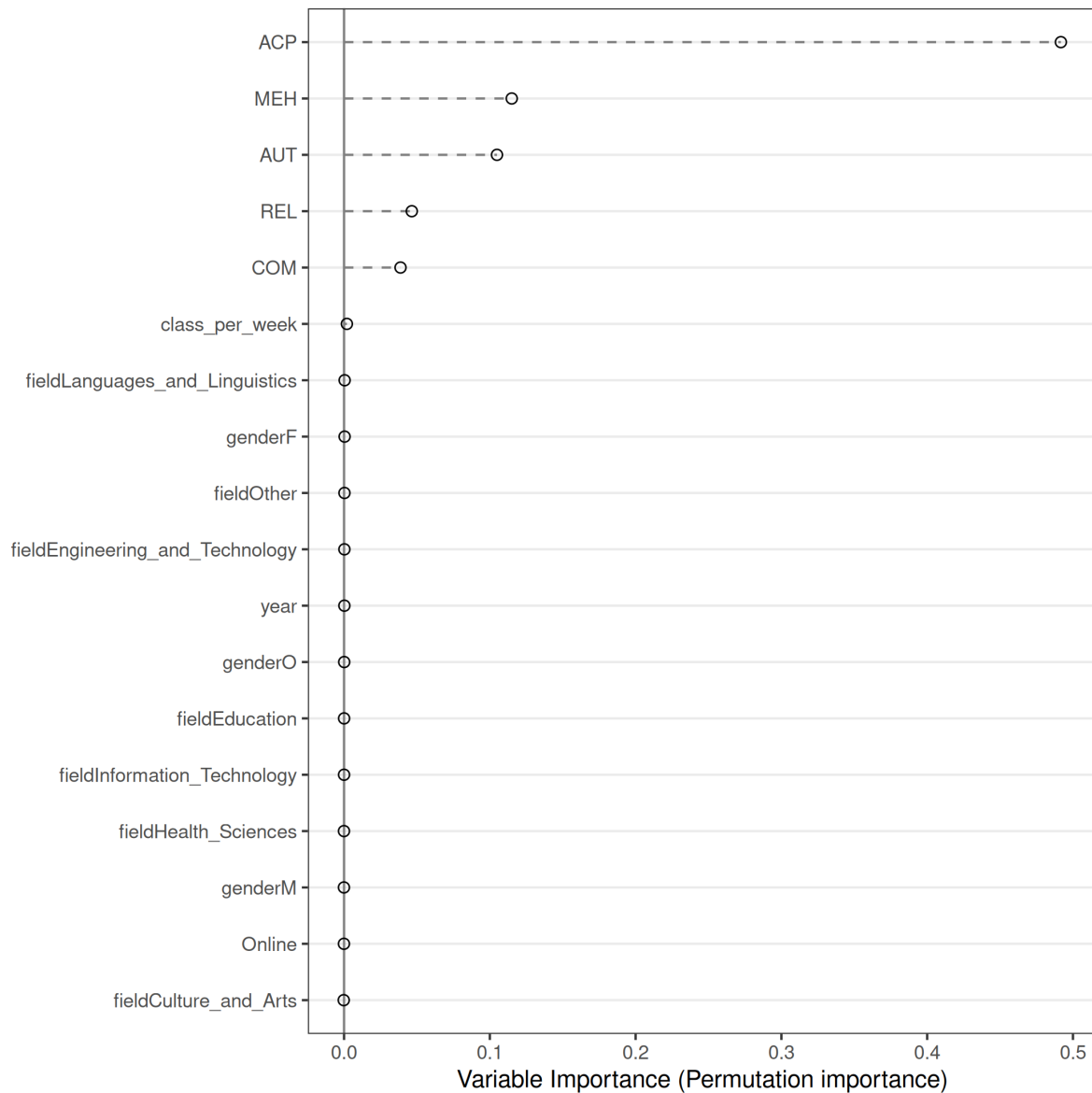
- which variables are selected at all; these are relevant predictors (ACP, AUT, REL, and MEH)

- how the model comes to a prediction; for example, if you have a score of 0.50 on ACP, and a score of 1.00 on MEH, then you end up in the right-most leaf of the tree, with a predicted value of 4.8 on ACM.
- Note that the previous example also illustrates a potential interaction between ACP and MEH because you first split on the one, and then on the other.

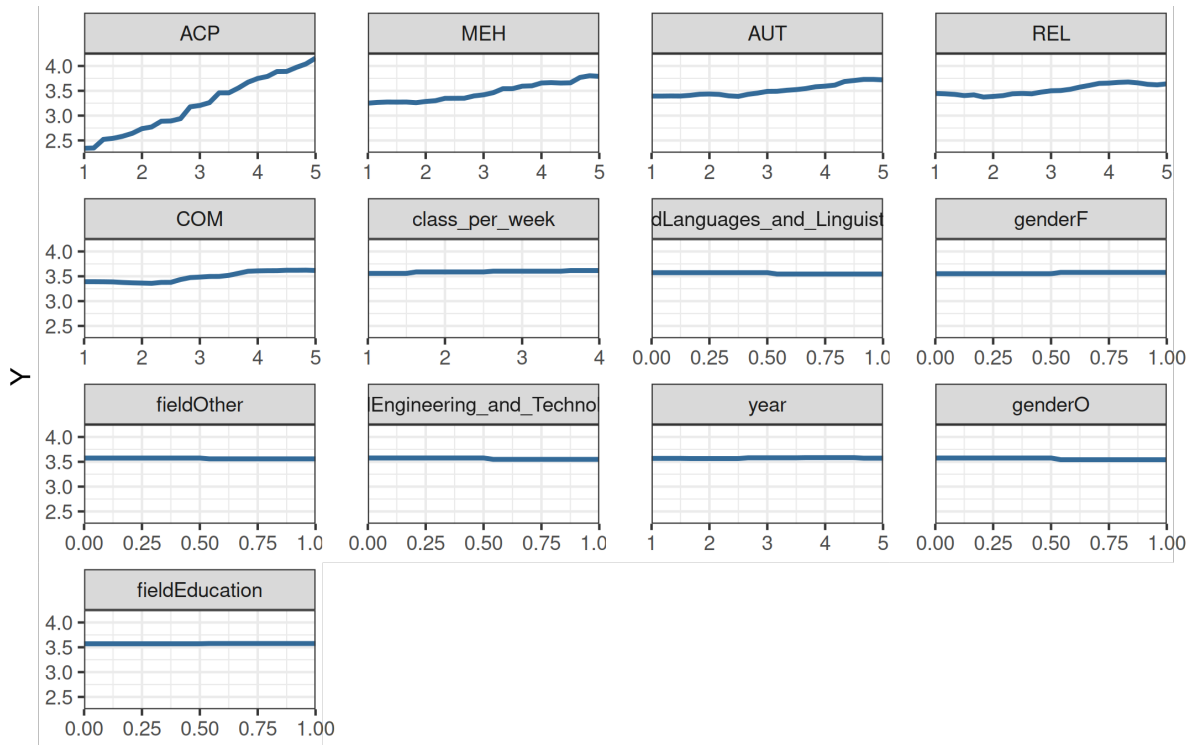
6.2.4.4 Explain a Random Forest

Unlike a single tree, a random forest is not interpretable, but you can analyze/explain its predictions to some extent.

First, you can randomly shuffle each predictor, and see how much *worse* the model predicts when that happens. If shuffling one predictor really diminishes predictive performance, it must have been an important predictor. Looking at the plot below, we see that the order of predictor importances is about the same as for the LASSO coefficients, with ACP being the most important:



The above plot tells us which predictors matter; to get a sense of the shape of the relationship between each predictor and the outcome (averaging over all other predictors), we can use “partial dependence plots”:



We see a near-perfect linear relationship of ACP, and (approximately linear) monotonous positive relationships for all other predictors as well.

6.2.4.5 Interpret Symbolic Regression

Finally, interpreting the symbolic regression model is very straightforward; just examine the equation constructed by the algorithm:

$$3.6 + ACP$$

6.3 Homework

If you managed to run the analysis, then you should now have machine learning results that tell you something about the predictors of one variable that (loosely) corresponds to a specific concept in your DAG.

Interpret the results you've obtained, and reflect on your previous theory (the DAG). Do you see any predictors that are not in your DAG but should be? Did you learn anything new about the direction of effects, about interaction effects, about the shape of association between concepts in your DAG? Do any of the analyses have weird results that might have an

explanation totally unrelated to your theory? Write your reflections down to become part of your final report.

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Setup your Computer

This is a tutorial on how to set up your personal computer for the course. It guides you through the installation of several software packages. This vignette does not assume a prior installation of R, so it is suitable for novice users. You only have to perform these steps once for every computer you intend to use, and the entire process should take approximately 30 minutes if you start from scratch. In case some of the software is already installed on your system, you can skip those related steps.

Follow these steps in order:

1. Install [R \(free\)](#)
2. Install [‘RStudio’ Desktop \(Free\)](#)
3. Download the [RStudio project](#) for this course into a folder where you’d like to keep your working files for assignments.
4. Unzip the compressed folder, open it, and open `patterns_principles_exercises.Rproj` in RStudio.
5. Once RStudio has loaded, you’ll see a panel in the bottom right labeled “Files”. Click the file named `setup.R`. It will load. Select all code in this file, then either press [CONTROL+ENTER], or click the Run button.